A denoiser that also assigns energies

Protein Diffusion Models as Statistical Potentials

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ProteinEBM: one sequence-conditioned energy for scoring, mutation effects, and sampling.

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What remains after predicting a native structure?

A single structure does not tell us which alternative states are favorable, how mutations shift stability, or how a chain folds.

- Confidence scores assess a prediction; they do not define a conformational probability density.
- Standard diffusion models generate structures but generally do not expose a cheap scalar energy for arbitrary structures.
- Physical simulations expose energies and trajectories, but long timescales are expensive.

Question: can a transferable diffusion model learn a useful conformational energy directly?

Introduction; Sections 2.1–2.6. These are the motivating distinctions, not a claim that existing tools lack all dynamics information.

The core change: make the score a gradient

Ordinary denoiser

Network directly predicts a vector field:

$$s_\theta(x, s, t) \approx \nabla_x \log p_t(x \mid s).$$

Useful for reverse diffusion; the learned field need not be exactly conservative.

ProteinEBM

Network produces a scalar energy:

$$s_\theta(x, s, t) = -\nabla_x E_\theta(x, s, t).$$

Its gradient drives sampling; its scalar output ranks candidate structures.

At a fixed noise level and sequence, the target is

$$E_\theta(x, s, t) \simeq -\log p_t(x \mid s) + C(s, t).$$

Sections 2.3 and 3. Energy parameterization gives conservative forces; training requires second derivatives.

Define the object before interpreting its energy

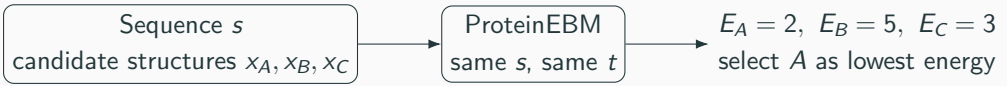
- s: amino-acid sequence. x: coarse-grained Cα coordinates, represented in nanometers during diffusion.
- t: diffusion noise level, not physical elapsed time. $E_\theta(x, s, t)$: dimensionless negative log-density target.
- A sequence-conditioned Boltzmann model has $p(x \mid s) \propto \exp[-\beta E(x, s)]$, with unknown normalizer Z_s.

Interpretation requires care

An energy difference between two coordinate points is not automatically a free-energy difference between two broad conformational basins. Basins require integrating over configurations.

Introduction; Section 3; our interpretation of the distinction between point energies and basin free energies.

A running example: one sequence, three candidate folds



If the learned energies equal negative log-density, then

$$\frac{p_t(x_A \mid s)}{p_t(x_B \mid s)} = e^{-(2-5)} = e^3 \approx 20.$$

The unknown normalizer cancels within the same sequence. This example illustrates the interface; the numbers are invented for teaching.

Conceptual example derived from the EBM formulation; not a paper result.

Training teaches the energy through denoising

For a native structure x_0, corrupt its Cα coordinates:

$$x_t = \sqrt{\alpha_t}\, x_0 + \sqrt{1-\alpha_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

Train $-\nabla_x E_\theta$ against the conditional Gaussian score

$$-\frac{x_t - \sqrt{\alpha_t} x_0}{1-\alpha_t}.$$

- Denoising score matching requires neither a partition function nor explicit negative structures.
- An auxiliary direct-score head accelerates some sampling; an all-atom head is an additional training task.
- The energy is still defined on Cα coordinates; auxiliary sidechain predictions are not an all-atom energy model.

Section 3; Appendix A. We use α_t for the paper's noise-schedule notation β_t to avoid confusion with thermodynamic β.

Architecture and data encode practical choices

- 85M parameters; downscaled AF3/Boltz-1 diffusion architecture. Rotations are learned through augmentation rather than guaranteed by equivariance.
- Pretraining pool: 32k CATH domains, 590k AFDB domains, 18k two-chain complexes. An epoch uses 32k AFDB examples alongside CATH and complexes.
- Selected models are finetuned on up to 3M MD frames from about 1k domains, simulated at 300 K.
- External-contact flags mark residues whose interaction partners were cropped away. At inference these flags are zeroed.

Why the flag matters: a standalone chain should not receive favorable scores by silently assuming a missing partner.

Section 3; Appendix A. MD finetuning and model variants depend on the downstream task.

Why an expert at low noise beats the full-range model

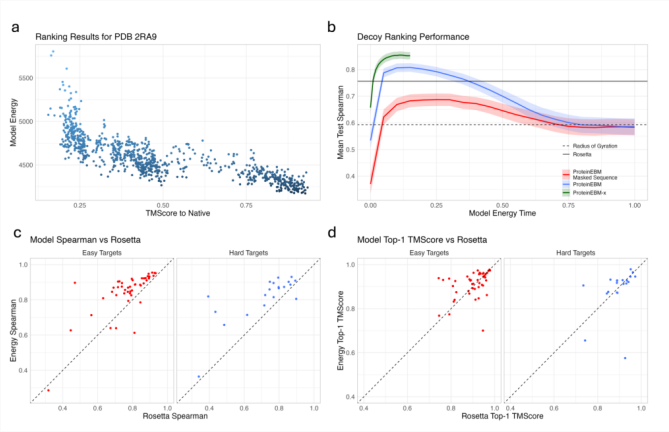
At extremely low noise, denoising mainly corrects local geometry. At very high noise, the density resembles a Gaussian and rewards compactness.

Variant	Training noise range	Main use
ProteinEBM	Full range $0 \leq t \leq 1$	Broad exploration
ProteinEBM-x	Concentrated below $t = .15$	Scoring and refinement

The default scoring level is $t = .05$, selected on a validation set. Thus the best empirical energy is a slightly smoothed density, not the formal zero-noise limit.

Section 4.1; Appendix A. Additional expert finetuning concentrates near $t = .05$.

Decoy ranking improves overall correlation



133 Rosetta proteins split approximately half validation/half test; training excludes $\geq 40\%$ sequence identity. Twenty test targets also have topology-level exclusion.

Mean ranking correlation magnitude: **.838 vs .757** for Rosetta ($p = .0005$).

Top-1 TMScore: **.905 vs .899**, $p = .46$; no significant gain in this endpoint.

Fig. 2; Section 4.1. Lower energy corresponds to higher TMScore; reported correlation values use the favorable ranking direction.

Mutation effects require a reference ensemble

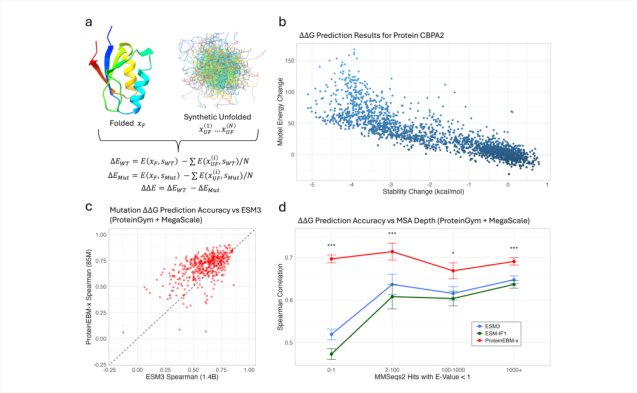
A sequence-dependent constant $C(s,t)$ makes raw cross-sequence energies unsafe to compare. Folded-minus-unfolded contrasts cancel that offset within each sequence.

$$D(s) = E(x_F, s, t) - \frac{1}{M}\sum_{j=1}^{M} E(x_{UF}^{(j)}, s, t), \qquad \Delta\Delta E = D(s_{WT}) - D(s_{Mut}).$$

- Input: wild-type folded backbone, wild-type/mutant sequences, synthetic unfolded backbones.
- The unfolded reference samples Ramachandran-random torsion angles.
- It assumes the folded and unfolded ensembles change little under mutation; it is a heuristic energy estimator, not full free-energy integration.

Section 4.2; Fig. 3a. Sign conventions differ across stability datasets; this follows the figure's WT-minus-mutant energy contrast.

Zero-shot stability ranking is the strongest result



ProteinGym mean Spearman: **.686** (Table 1), versus ProSST .653 and ESM3 .641.

Across ProteinGym plus additional MegaScale proteins, advantage is greatest for shallow evolutionary histories.

0–1 MMseqs2 hits: **.697 vs .519** for ESM3; difference .178 ($p < .0002$).

Fig. 3; Table 1; Section 4.2. Text reports .684 while Table 1 reports .686: manuscript inconsistency, not extra precision. Error bars in panel d are SEM.

Relative mutation ranking is not absolute stability

Absolute folding-stability estimator	Spearman	MAE (kcal/mol)
ProteinEBM-x	.471	1.06
ESM-IF comparison from Cagiada et al.	.672	.96
Sequence-length baseline	.415	1.12

The absolute estimator adds fitted linear coefficients and a sequence-length entropy term. Even then it performs only modestly above length alone. Takeaway: strong mutation ranking does not establish calibrated folding free energies across different proteins. The unfolded-state approximation is a plausible bottleneck.

Appendix E; Fig. S4. The unfolded-state explanation is the authors' hypothesis.

Two sampling modes answer different questions

Langevin annealing

Start from Gaussian noise; alternate diffusion denoising with local Langevin steps. Use the base model above $t = .1$ and the expert below.

Good for broad structural search and selecting low-energy conformations. Noise-to-structure paths are not physical folding paths.

Fixed-noise Langevin dynamics

Start from an unfolded chain and remain on the energy surface at $t = .05$:

$$x_{k+1} = x_k - h\nabla E(x_k) + \sqrt{2h}\,\eta_k.$$

Useful for local transitions and qualitative folding pathways; simulation steps lack calibrated physical time.

Sections 4.3–4.4; Appendix B. Simplified dimensionless Langevin update absorbs the friction coefficient into h.

Fast folders: native basins emerge, populations remain uncertain

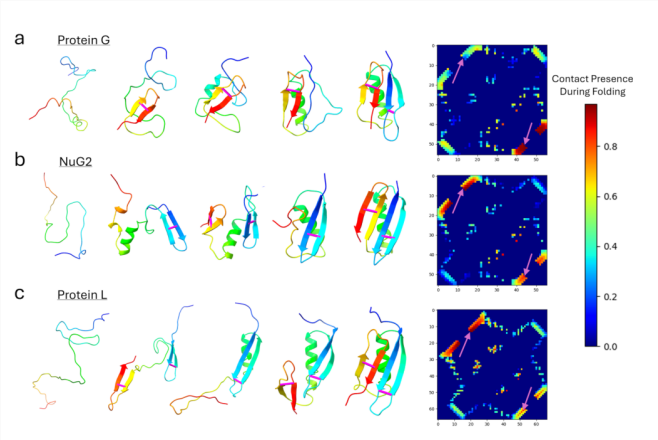
Eleven fast-folding proteins were excluded from training at 40% sequence identity.

- Every target had at least one sample within 2.5 Å RMSD of native.
- The lowest-energy sample was within 3.5 Å for 10 of 11 targets. WWDomain retained the topology but differed in loops.
- NuG2 sometimes prefers a strand-swapped basin; low energy is not a guarantee of native topology.
- TICA projections overlap MD landscapes qualitatively, but sample less non-native space. Training MD was at 300 K; reference MD was at 350 K.

Native-state discovery is clearer evidence here than accurate equilibrium populations.

Fig. 4; Section 4.3; Fig. S2; Appendix B.2. Our interpretation in the last line.

Folding pathways recover a known sequence-dependent switch



Same overall topology; different early hairpin formation.

Protein G: C-terminal contacts dominate transition paths.

NuG2 and Protein L: shift toward N-terminal contacts, consistent with experimental pathway evidence.

Contact presence fraction is a pathway summary, not an experimental rate constant.

Fig. 5; Section 4.4. NuG2 was engineered from Protein G to stabilize the N-terminal hairpin.

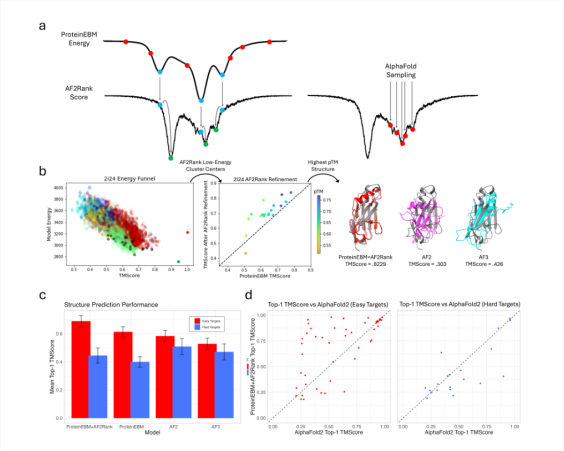
Successful folding paths are a selected minority

Protein	Native-reaching trajectories	Approximate fraction
Protein G	22 / 1200	1.8%
NuG2	34 / 800	4.3%
Protein L	25 / 4800	0.5%

Native-reaching means RMSD < 3 Å with correct secondary-structure pairings. The pathway analysis uses trajectories reaching native. Protein L additionally uses a weak compaction potential; its lowest-energy sampled structure has swapped strands. Coarse-graining can alter kinetics.

Section 4.4; Appendices B.2 and D; Fig. S3. The observed fractions are not physical folding probabilities or calibrated rates.

Prediction separates exploration from final selection



1. Anneal with MD-finetuned base model.
2. Resample and refine with the expert.
3. Optionally send ≤ 40 cluster representatives to AF2Rank.
4. Select its highest-pTM output.

Single-sequence baselines: 10k AF2 and 15k AF3 samples per target.

Fig. 6; Section 4.5; Appendices B.2 and H. Baseline counts are matched by empirical runtime estimates, not identical hardware execution graphs.

The benefit depends on what is held out

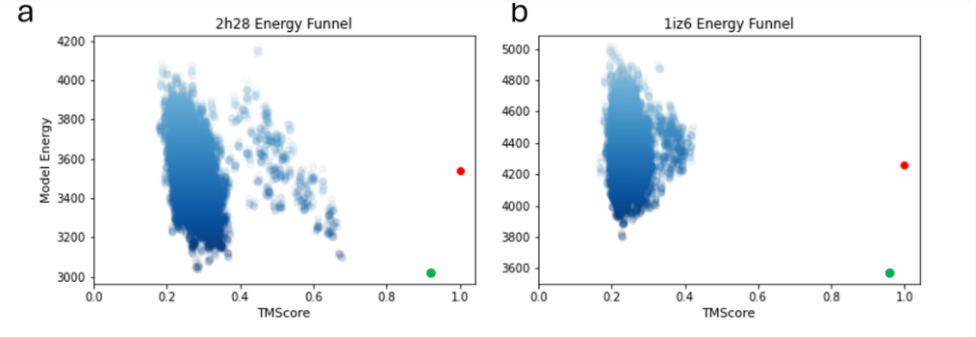
Method	Easy targets: mean top-1 TMScore	Comparison to AF2
AF2, single sequence	.584	—
AF3, single sequence	.529	—
ProteinEBM	.613	$p = .43$
ProteinEBM + AF2Rank	.690	$p = .001$

Easy: sequence-held-out for ProteinEBM; not necessarily excluded from AlphaFold training. Hybrid inference gives the strongest result.

Hard: topology-held-out. ProteinEBM underperforms AF2 and AF3 even after AF2Rank refinement, despite retaining good decoy ranking.

Fig. 6c–d; Section 4.5. Easy/hard refer to the ProteinEBM training split, not a universal difficulty classification.

Diagnose failure before spending more sampling compute



Ranking failure: native-like samples exist, but a worse structure receives lower energy. Better sampling alone cannot fix the final choice.

Exploration failure: the native has favorable energy, but the sampler never reaches it. Improve proposals or enhanced sampling.

Fig. S9; Appendix J. Red = native structure, green = relaxed native; TMScore on x-axis, model energy on y-axis.

What would we trust it to do today?

- **Supported**: rank mutations and plausible structures; use an explicit energy to guide and diagnose structural search.
- **Promising**: screen designed backbones cheaply. On 500 Boltzgen designs, AUC=.952 for Boltz-2 pTM $>$ .7 and .889 for refolding RMSD $<$ 2 Å. These are predictor-based labels, not experimental success.
- **Not established**: absolute state populations, kinetic rate constants, broadly superior prediction of unseen topologies, or accurate all-atom energetics.

Discussion: for our use case, is the main bottleneck ranking, exploration, or calibration? Which independent experiment would distinguish these?

Main results; Appendices E–G and J. Code/weights: github.com/jproney/ProteinEBM. LplA occupancy analysis uses activity as a proxy and trails ProteinMPNN.